Sicheng Wang

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Experience

ByteDance **San Jose, CA**

Software Engineer *Oct. 2024 — Present*

- Designed and built a permission-management **AI agent** for Lark (ByteDance's Slack equivalent), from 0 to 1, scaling to **2,000** active users and **10,000+** conversations.
- Built an oncall-ticket summarization pipeline feeding the agent's knowledge base, enabling it to answer user questions from historical oncall summaries and cutting weekly oncall tickets from ∼160 to ∼100 (**30%+** reduction).
- Root-caused the agent's weak oncall Q&A accuracy to imprecise RAG retrieval, widened recall and reranked with a stronger model — lifting answer accuracy by **37.5%**.
- Built an MCP server helping agents search the permission-management platform's knowledge base and cross-service logs to self-diagnose permission errors, ranking **#5**/230 on ByteDance's MCP marketplace.
- Drove cross-team development of a Data Analytics Agent on Hive data with the data team, meeting **90%+** of the operations team's data-analysis needs.
- Full-stack developed an **ABAC**-based auto-approval system spanning the admin console, rule engine, and backend APIs, increasing company-wide auto-approval rates and reducing time-to-approval by **49.3% YoY**, saving an estimated **310,000+** person-days in Q4 2025 alone.
- Optimized multithreaded processing logic of the permission-reclamation detail API, cutting response time by **57%**.
- Built a pre-production test environment and automated deployment pipelines for TikTok's US Data Security (USDS) organization, eliminating environment-related production rollbacks (down from ∼2/year to zero).

Stackline **Seattle, WA**

Software Development Engineer II *Mar. 2024 — Sept. 2024*

Software Development Engineer I *Apr. 2023 — Mar. 2024*

Software Development Intern *Jun. 2022 — Sept. 2022*

- Architected and implemented scalable web crawling microservices to collect data from **30+** retailers, efficiently processing **60 million+** requests weekly and supporting the company's analytical products.
- Automated **70+** crawling pipelines using **Apache Airflow** and leveraged **AWS Kinesis Data Firehose** to stream results to **AWS Redshift** and **S3**, facilitating seamless collaboration with data science team's analytics workflows.
- Created a cost-efficient serverless data reprocessing service with **AWS Lambda** for Walmart FPR crawl jobs, eliminating redundant web requests and reducing operational costs by **44%**.
- Developed and integrated a CAPTCHA solver using **Puppeteer** screenshots and image analysis, with Redis-based locking to coordinate synchronized execution across multiple browser instances and ECS tasks.
- Developed an Amazon Deals crawling pipeline and deployed it on **AWS ECS** with **GitLab CI/CD** to collect promotional information of **19** countries, crawling over **10 millions** deals per week.
- Reduced crawling overheads by **20%** through deducing deal product variation status and removing redundant crawls.

Publications

Long-standing research background in NLP/LM; selected publications below (Google Scholar).

- **NeighXLM: Enhancing Cross-Lingual Transfer in Low-Resource Languages via Neighbor-Augmented Contrastive Pretraining**. *Findings of the Association for Computational Linguistics: EMNLP 2025*. Sicheng Wang, Wenyi Wu, Zibo Zhang. (1 citation)
- **CERT: Contrastive Self-Supervised Learning for Language Understanding**. *arXiv preprint*, 2020. Hongchao Fang, Sicheng Wang, Meng Zhou, Jiayuan Ding, Pengtao Xie. (**469** citations, highly cited)

Skills

- AI & Agents: Agent Development, RAG, MCP, LangChain/LangGraph, Agent2Agent (A2A), PyTorch
- Languages: Go, Python, TypeScript/JavaScript, C/C++, Haskell
- Frameworks and Tools: React, Node.js, RESTful API, AWS, Docker, Kubernetes, Elasticsearch, Redis, MongoDB, SQL, Airflow, Spark, Git, CI/CD

Education

University of California San Diego **San Diego, CA**

- Master of Science in Computer Science, GPA: 3.87/4.00 *Sept. 2021 — Mar. 2023*

Tsinghua University **Beijing, China**

- Bachelor of Engineering in Automation, GPA: 3.83/4.00 *Aug. 2017 — Jul. 2021*
- Awards: Philobiblon Scholarship (**Top 5%**), 2019 — Guangyao Scholarship (**Top 10%**), 2018